

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

**20BTC211 - PLANT DESIGN AND ECONOMICS**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

**Unit – I****Marks BT\* CO\* PO\***

- |       |                                                                                                                                                                              |    |     |   |      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|---|------|
| 1. a) | The geographical location of the final plant can have strong influence on the success of an industrial venture. How do you justify this considering valid factors?           | 10 | L*1 | 1 | 1    |
| b)    | Highlight on the technical and equipment related factors which are critical for process selection based on comparisons.                                                      | 10 | L3  | 2 | 3    |
| 2. a) | List the different types of project and mention the steps for development of an industrial plant from grass root level.                                                      | 10 | L1  | 1 | 1    |
| b)    | Briefly describe the different types of flow diagrams that are available and create a qualitative flow diagram for the ammonia production plant utilising Haber's technique. | 10 | L3  | 2 | 3    |
| 3. a) | Briefly explain the possible options for effective management of solid waste management and particulate air pollutants.                                                      | 10 | L2  | 1 | 1,11 |
| b)    | What is process design? Write a brief note on types of process design.                                                                                                       | 10 | L1  | 2 | 2    |

**Unit – II**

- |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |    |   |   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 4. a) | Explain different types of capital cost estimates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10 | L2 | 3 | 4 |
| b)    | Define:<br>i) Service life    ii) Salvage value    iii) Present value<br>iv) Book value    v) Market value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10 | L1 | 4 | 1 |
| 5. a) | A reactor, which will contain corrosive liquids, has been designed. If the reactor is made of mild steel, the initial installed cost will be \$5000 and the useful-life period will be 3 years. Since stainless steel is highly resistant to the corrosive action of the liquids, stainless steel, as the material of construction, has been proposed as an alternative to mild steel. The stainless-steel reactor would have an initial installed cost of \$15,000. The scrap value at the end of the useful life would be zero for either type of reactor, and both could be replaced at a cost equal to the original price. On the basis of equal capitalized costs for both types of reactors, what should be the useful-life period for the stainless-steel reactor if money is worth 6 percent compounded annually? | 10 | L4 | 3 | 2 |
| b)    | Explain any two methods for determining depreciation which does not consider interest.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 | L1 | 4 | 4 |
| 6. a) | Make a study estimate of the FCI for a process plant if the purchased equipment cost is 10000\$. Use the ranges of process plant component cost outlined in the table below for the process plant handling both solids and fluids with a high degree of automatic controls and essentially outdoor operation. Do not include land.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |   |   |

**P.T.O.****10 L4 3 4**

| Component                        | Selected percentage of FCI |
|----------------------------------|----------------------------|
| Purchased equipment              | 25                         |
| Purchased equipment installation | 9                          |
| Instrumentation (installed)      | 10                         |
| Piping (installed)               | 8                          |
| Electrical (installed)           | 5                          |
| Buildings (including services)   | 5                          |
| Yard improvement                 | 2                          |
| Service facilities (installed)   | 15                         |
| Engineering and supervision      | 8                          |
| Construction expense             | 10                         |
| Legal expense                    | 2                          |
| Contractors fee                  | 2                          |
| Contingency                      | 8                          |
| Total                            | 109                        |

- b) An industry has purchased a fermenter for \$20,000, completely installed and ready for operation. Its salvage value is estimated to be \$2000 at the end of a service life estimated to be 5 years. Find the yearly depreciation value and asset values for all the years. Solve this problem by sum of years digit method.

10 L4 4 2

## Unit – III

7. a) A new manufacturing unit has just been constructed and put into operation by your company. The basis of the manufacturing process is a special computer for control (designated as OVT computer) as developed by your research department. The plant has now been in operation for less than one week and is performing according to expectations. A new computer (designated as NTR computer) has just become available on the market. This new computer can easily be installed at once in place of your present computer and will do the identical job at far less annual cash expense because of reduced maintenance and personnel costs. However, if the new computer is installed, your present computer is essentially worthless because you have no other use for it. Following is pertinent economic information relative to the two computers:

Assume that the after tax savings is based on 35% tax rate

|                                       | OVT computer | NTR computer |
|---------------------------------------|--------------|--------------|
| Capital investment                    | 2000000\$    | 1000000\$    |
| Estimated economic life               | 10 yrs       | 10 yrs       |
| Salvage value at end of economic life | 0            | 0            |
| Annual cash expenses                  | 250000\$     | 50000\$      |

What recommendation would you make relative to replacing the present \$2,000,000 computer with the new computer?

10 L4 5 2

- b) Define profitability and explain any two methods of profitability evaluation.

10 L2 5 4

8. a) Explain briefly the following terms:  
 i) Payout period      ii) Capitalized cost      iii) Book value  
 iv) Replacement      v) Alternative investment  
 b) Briefly discuss the organization of an oral report.

10 L2 5 2

10 L2 5 4

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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